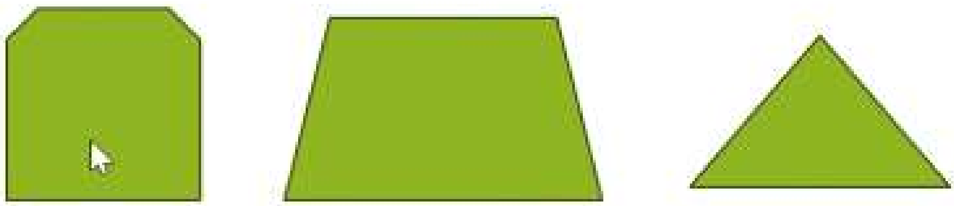
# Polygon

- In computer graphics, a polygon is a closed, two-dimensional area primitive defined by a series of straight-line segments (edges) connected by vertices.
- In computer graphics, a polygon is a fundamental graphics primitive defined as a closed planar figure bounded by a finite set of straight line segments called edges.
- Types of Polygons : Polygons are primarily classified based on their interior angles and how they occupy space:
  - **Convex Polygons:** All interior angles are 180 degree or less. A line segment connecting any two points inside the polygon will always lie entirely within it.



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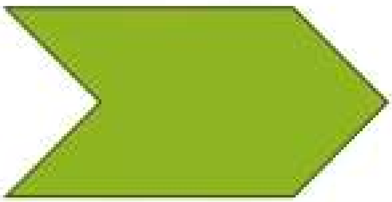
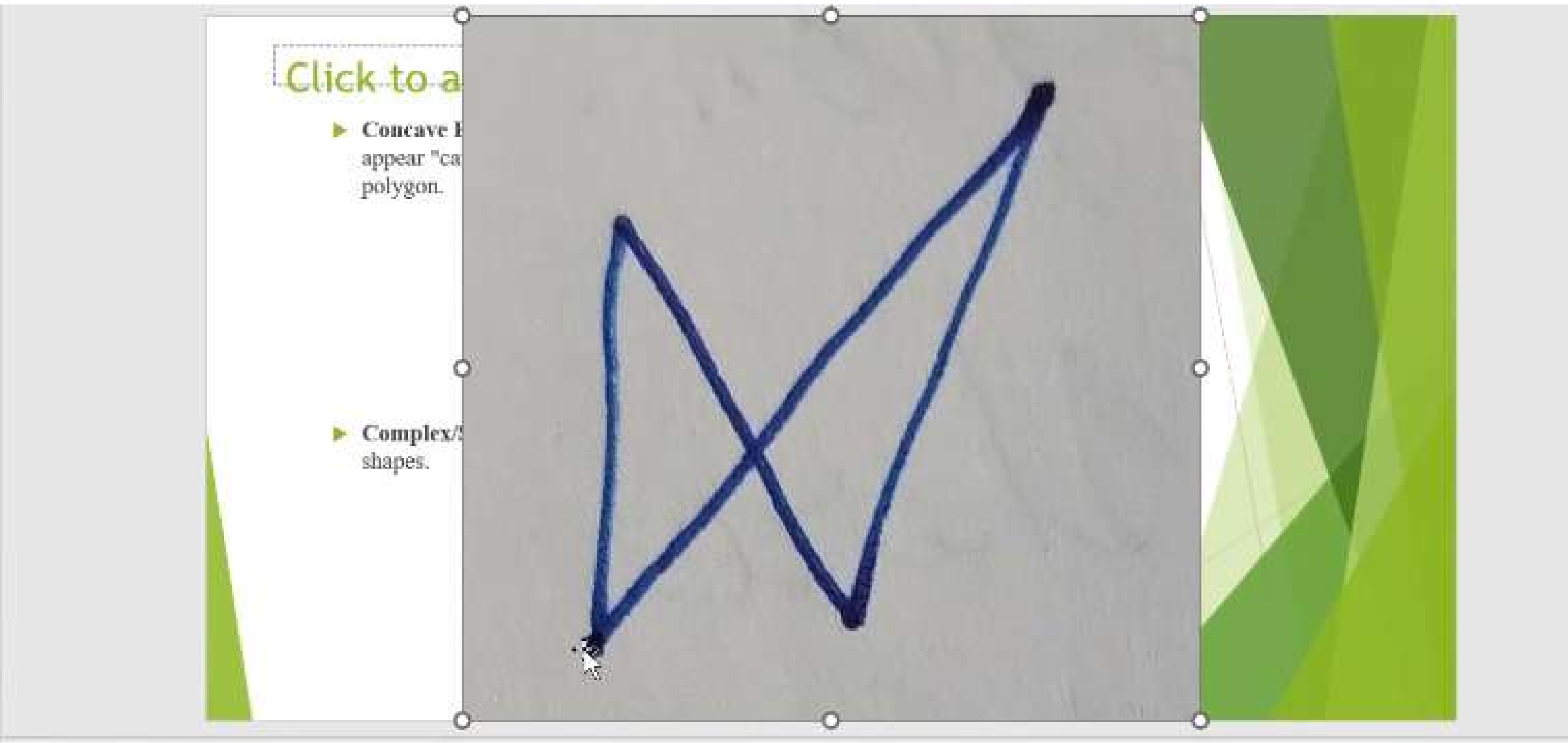
- **Concave Polygons:** At least one interior angle is greater than 180 degree.These appear "caved in," and a line connecting two interior points may pass outside the polygon.



- **Complex/Self-Intersecting:** Polygons where edges cross each other, such as star shapes.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Complex/
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# Polygon Filling Algorithm

- Polygon filling is a process of coloring each and every pixel of that comes inside the polygon region.
- There are two basic approaches used for polygon filling:
- Seed Fill Algo.
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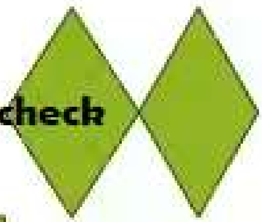
    - Inside Outside Test
        - 1) Odd even rule **/even-odd / odd parity check**
        - 2) Non zero winding rule



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